Phenomenological Regimes Induced by Structural Identity

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Abstract

We study structural assignments from inverse-limit histories into an abstract metric codomain. Exact equality of all finite readouts forces factorization through the first cut and is incompatible with the required non-factorization. We replace that equality by uniformly summable readout increments, retaining non-factorization as a separately checked condition.

An explicit binary-prefix system with ternary readouts realizes both conditions. It gives uniform approximation, a global lower and upper readout bound, and an exact minimax error for each finite observation budget. Continuity follows from controlled approximation by continuous finite readouts. Cross-system fragmentation and invariance require typed comparison domains and audited transports; a scalar rigidity label alone does not imply them. Under those comparison hypotheses we derive loss and normalized protection statements. The abstract codomain is not identified with consciousness, subjective experience, or moral status.

1 Scope, System Boundary, and Non-Inference Note

Scope and admissible reading. This paper defines an abstract regime codomain and studies controlled assignments, a concrete realization, and conditional fragmentation/invariance results using explicit comparison data in addition to upstream identity structure. The results are conditional statements about mathematical compatibility. They do not assert that phenomenology exists, that CCR systems are conscious, that biology or computation instantiate the framework, or that identity structure causes experience. Related runtime diagnostics may instantiate continuity-side or anti-fragmentation checks downstream, but they do not implement phenomenology and do not externally validate the regime classification. No runtime output, continuity proxy, or internal support artifact should be read as proof that phenomenology is present or that the structural-ethics layer has empirical closure.

2 Introduction

2.1 Background and Motivation

Recent work [2] established a formal framework for rigid identity as an inverse limit in observable channels. That framework classifies systems by a scalar invariant, the ontological mass M_Ω, which quantifies resistance to identity deformation under perturbation. Three regimes emerge:

- $M_\Omega = 0$: zero deformation-cost infimum;
- $0 < M_\Omega < \infty$: deformable persistence (finite rigidity);
- $M_\Omega = +\infty$: closed–rigid (CCR rigidity).

That framework is ontologically silent regarding phenomenology. It characterizes identity structure, not experience. Connecting its rigidity regimes to assignment behavior requires the explicit readout and comparison hypotheses studied below.

This paper addresses the following formal question:

Given a regime of structural identity rigidity, what classes of phenomenological dynamics are mathematically compatible or incompatible with it?

All results are conditional: they state what is *structurally compatible or incompatible* under given assumptions, not what *is the case*.

2.2 Central Results and the Remaining Rigidity Link

Controlled asymptotic compatibility admits a nonempty realization with no finite-cut determination. The regularity bounds and finite-budget lower bound are derived for that realization. Continuity and invariance are distinct conclusions: the first follows from regular finite readouts and summable tails, while the second requires a transport-compatible readout comparison.

The broader rigidity program asks when finite and infinite deformation-cost regimes enforce the respective comparison branches. This paper states and uses those hypotheses explicitly; it does not infer them from the scalar alone. Establishing them on larger classes is retained as a research obligation.

2.3 Ontological Neutrality

This work maintains strict ontological neutrality:

1. We define mathematical objects (phenomenological spaces, compatibility operators).
2. We derive structural constraints from axioms.
3. We state impossibility theorems.
4. We construct mathematical realizations, without asserting their physical or experiential instantiation.

The framework is therefore a theory of *structural possibility*, not a theory of *phenomenal reality*.

2.4 Terminological Alignment: Structural Regime Assignments

The phrase "phenomenological regime" is retained for continuity with the published corpus, but the demonstrated object in this paper is more precisely a *structural regime assignment*. The mathematical content is a constrained map from identity structure into an abstract codomain. No semantic clause in this paper identifies that codomain with experience.

Definition 2.1 (Structural regime alias)**.** Whenever the paper writes a phenomenological space $\mathcal{E}$ or a phenomenological assignment $\Phi : \mathcal{I} \to \mathcal{E}$, the strictly demonstrated reading is:

$$\mathcal{E}_{\mathrm{str}} := \mathcal{E}, \qquad \Phi_{\mathrm{str}} := \Phi,$$

where $\mathcal{E}_{\mathrm{str}}$ is an abstract structural regime codomain and Φ_{str} is a compatibility assignment into that codomain.

Proposition 2.2 (No identification with experience)**.** *The formal apparatus of this paper does not define a map from $\mathcal{E}_{\mathrm{str}}$ to human experience, phenomenal consciousness, qualia, or subjective reports.*

Proof. Definitions in this paper specify an abstract codomain, compatibility conditions, and rigidity-dependent constraints. They do not introduce a measurement channel from $\mathcal{E}_{\mathrm{str}}$ to biological reports, human psychophysics, phenomenal properties, or moral status. Therefore no such identification is present in the formal language of the paper. □

2.5 Relation to BaseCore

This paper is the standalone expository source for the regime-assignment development. The current BaseCore package also contains an absorbed, dependency-facing version of the formal core where those definitions and theorems function as base-layer exports for later papers. The two surfaces therefore have different editorial roles: Paper 2 gives the full local derivation and interpretation boundary, while BaseCore supplies the normalized theorem-export surface used by the corpus.

3 Formal Setup

3.1 Review of Identity Framework

We assume the identity framework of [2]. Let $(S_t, \pi_{t+1\to t})_{t\in\mathbb{N}}$ be a projective system of observable state spaces. The identity object is the inverse limit:

$$\mathcal{I} := \varprojlim S_t = \left\{ (x_t)_t \in \prod_t S_t \;\middle|\; \pi_{t+1\to t}(x_{t+1}) = x_t \;\; \forall t \right\}. \tag{1}$$

The ontological mass M_Ω quantifies resistance to identity deformation. We recall that:

- $M_\Omega = 0$: zero deformation-cost infimum;
- $0 < M_\Omega < \infty$: deformable persistence (finite rigidity);
- $M_\Omega = +\infty$: closed–rigid (CCR rigidity).

3.2 Abstract Phenomenological Space

Definition 3.1 (Phenomenological Space)**.** A phenomenological space is a nonempty set $\mathcal{E}$ whose elements are called *phenomenological configurations*. No topology, metric, or algebraic structure is assumed a priori.

Remark 3.2. We do not claim that $\mathcal{E}$ represents "conscious states" or "qualia". It is an abstract mathematical space whose interpretation is not specified by the formalism. The space $\mathcal{E}$ is introduced purely as the codomain of a structural assignment; its ontological status is deliberately left open.

3.2.1 Why a Separate Space?

One might ask: why not use $\mathcal{I}$ itself as the phenomenological space? The answer is structural:

Proposition 3.3 (Endomorphisms Preserving All Canonical Projections)**.** *If $F : \mathcal{I} \to \mathcal{I}$ satisfies $p_t \circ F = p_t$ for every t, then $F = \mathrm{Id}$. This does not exclude endomorphisms with nonidentity coordinate actions.*

Proof. For each x, $F(x)$ and x have identical coordinates, hence are the same inverse-limit element. For contrast, on binary prefix spaces, bitwise complement defines an endomorphism induced by complement on every prefix; it commutes with the bonding maps but does not preserve each coordinate value. A separate codomain is a modeling choice, not a logical necessity for nonidentity maps. □

A separate codomain permits independent structural readouts; it is not forced by the universal property alone.

3.3 Phenomenological Assignment

Definition 3.4 (Phenomenological Assignment)**.** A phenomenological assignment is a family of total maps on the declared observable state spaces:

$$\phi_t : S_t \to \mathcal{E} \tag{2}$$

indexed by $t \in \mathbb{N}$.

Definition 3.5 (Induced Limit Assignment)**.** Given a phenomenological assignment $\{\phi_t\}$, the induced limit assignment, whenever it exists, is the map:

$$\Phi : \mathcal{I} \to \mathcal{E}, \quad \Phi((x_t)_t) := \lim_{t\to\infty} \phi_t(x_t), \tag{3}$$

The convergence statements below use the complete metric codomain of Hypothesis 4.3. Partial readouts may instead be studied on a specified common domain, with all projections and factorizations restricted to that domain.

3.3.1 Uniqueness of the Limit Assignment

Proposition 3.6 (Uniqueness and Uniform Error of the Limit Assignment)**.** *Under E1, each trajectory has a unique limit assignment and*

$$d_{\mathcal{E}}\big(\Phi(x), \phi_n(p_n x)\big) \le A_n \quad \textit{for all } x \in \mathcal{I}.$$

Proof. For $m > n$, the triangle inequality bounds the distance between the mth and nth readings by $\sum_{t=n}^{m-1} a_t$. Thus each sequence is Cauchy. Completeness supplies its limit, uniqueness follows from metric separation, and passage to the limit gives the bound. Since $A_n \to 0$ independently of x, convergence is uniform. □

3.4 Axioms for Phenomenological Coherence

We declare a controlled assignment model with explicit logical dependencies and a constructive nonempty realization.

Axiom 3.7 (E0: Non-Triviality)**.** The phenomenological space satisfies $|\mathcal{E}| \ge 2$, i.e., there exist at least two distinct phenomenological configurations.

Axiom 3.8 (E1: Controlled Asymptotic Compatibility)**.** In the complete metric codomain of Hypothesis 4.3, there are numbers $a_t \ge 0$, independent of the trajectory, with $\sum_{t=0}^{\infty} a_t < \infty$, such that

$$d_{\mathcal{E}}\big(\phi_{t+1}(p_{t+1}x), \phi_t(p_t x)\big) \le a_t \qquad (x \in \mathcal{I},\ t \ge 0).$$

Here $p_t : \mathcal{I} \to S_t$ is the canonical projection. Define the tail budget $A_n = \sum_{t=n}^{\infty} a_t$. Exact equality is the special case $a_t = 0$, not a requirement of E1.

Axiom 3.9 (E2: Non-Locality)**.** There exists no function $f : S_t \to \mathcal{E}$ (for any fixed t) such that for all trajectories $(x_t)_t \in \mathcal{I}$:

$$\Phi((x_t)_t) = f(x_t). \tag{4}$$

3.5 Status of the Assignment Conditions

The following results separate sufficient assumptions, a non-factorization requirement and their logical dependencies.

3.5.1 Nontriviality of the Codomain

Theorem 3.10 (E0 is Necessary). *If $|\mathcal{E}| = 1$, then every phenomenological assignment is constant and no phenomenological classification is possible.*

Proof. If $\mathcal{E} = \{e_0\}$, any total assignment takes the value e_0 on every trajectory. It distinguishes no trajectories and factors through every canonical projection. It therefore cannot meet E2 on a nonempty domain. □

3.5.2 Controlled Convergence

Theorem 3.11 (Controlled Compatibility Is Sufficient). *E1 guarantees existence of the limit with the error bound in Proposition 3.6. Absolute summability is sufficient, not necessary for an arbitrary convergent family.*

Proof. Existence and the bound were proved above. To disprove necessity in general, take readouts constant in the trajectory with values $s_n = \sum_{k=1}^{n} (-1)^{k+1}/k$ in $[0, 1]$. The even partial sums increase, the odd partial sums decrease, and their difference tends to zero, so they have a common limit. Their successive absolute increments form the divergent harmonic series; no summable uniform majorant exists. This example is a control for the necessity claim, not a realization of E2. □

3.5.3 Finite-Cut Non-Factorization

Theorem 3.12 (Non-Factorization Is an Independent Requirement). *E2 excludes readouts determined by a single canonical finite cut. The inverse-limit construction and E1 alone do not imply E2.*

Proof. Take any nonempty inverse limit and a constant family $\phi_t = e_0$ in a codomain containing at least two points. E1 holds with zero tail, but $\Phi = e_0$ factors through every projection. Thus E2 must be checked separately. The nonempty construction in Theorem 3.15 checks it without assuming it. □

3.6 Logical Dependencies and Realizability

Theorem 3.13 (Logical Dependencies of the Assignment Conditions). *On a nonempty trajectory domain, E2 implies E0. E1 does not imply E2. There exist assignments satisfying E0, the revised E1 and E2 simultaneously. Consequently these three conditions are not an independent minimal axiom basis as originally stated.*

Proof. If the codomain were a singleton, every assignment would be constant and factor through p_0, contradicting E2. The preceding constant-family control proves the second assertion. Theorem 3.15 proves the third by construction. E0 is retained as an explicit convention, although E2 entails it in the nonempty case. □

3.7 Rejected Alternatives

We briefly discuss axiom candidates that were considered and rejected.

3.7.1 Rejected: Continuity Axiom

One might require Φ to be continuous. This is not required by the chosen model because:

- The complete-metric model permits discontinuous finite readouts; continuity is a property of the regular-readout subclass, not part of the unrestricted assignment condition;
- Continuity is derived from controlled approximation by continuous finite readouts; a scalar CCR label alone is insufficient.

3.7.2 Rejected: Surjectivity Axiom

One might require Φ to be surjective. This is not required by the chosen model because:

- A readout need not exhaust a codomain chosen to include other regimes;
- Surjectivity is a strong structural claim not derivable from identity constraints.

3.7.3 Rejected: Injectivity Axiom

One might require Φ to be injective. This is not required by the chosen model because:

- Many trajectories may share phenomenological configurations;
- Injectivity is not imposed on the full assignment class; it is a consequence in the concrete coercive realization. An injective map alone is not necessarily a topological embedding.

3.8 Structural Interpretation

Axiom	Structural Meaning	Status
E0	Nontrivial codomain	E2 consequence
E1	Summable increments	Uniform limit
E2	Finite-cut non-factorization	Required

3.9 Closure of Section 2

The limit is uniquely defined with a uniform tail bound. E2 remains a substantive non-factorization requirement; E0 is redundant when E2 holds on a nonempty domain. Exact compatibility cannot coexist with E2, whereas the controlled condition has the explicit realization below. No existence or minimality claim is obtained merely by naming an axiom.

3.10 Phenomenological Neutrality Clause

Throughout this paper, the phenomenological space $\mathcal{E}$ is treated as a *purely abstract topological and metric structure*. No semantic, psychological, biological, or experiential interpretation is assumed or required for any of the formal results derived herein.

The term "phenomenological" refers exclusively to the role of $\mathcal{E}$ as an abstract codomain of structural compatibility for inverse-limit identities, and does not presuppose the existence of consciousness, subjective experience, or any particular phenomenological content.

All theorems, constructions, and stability results presented in this work are therefore statements about abstract mathematical objects only. Any interpretation of $\mathcal{E}$ in relation to experiential, cognitive, or phenomenological phenomena lies strictly outside the formal scope of this paper.

3.11 Exact Compatibility Obstruction

Proposition 3.14 (Exact Compatibility Forces Finite-Cut Factorization)**.** *On a nonempty trajectory domain, if $\phi_{t+1}(p_{t+1}x) = \phi_t(p_t x)$ for every x, t, then the limit assignment satisfies $\Phi = \phi_0 \circ p_0$. It cannot satisfy E2. The controlled E1 above replaces that exact equality, rather than assuming it together with E2.*

Proof. Induction makes all readings equal to the first. Their limit is therefore $\phi_0(p_0 x)$. For partial readouts with a declared common domain, the same argument holds on that domain. If a total map on S_0 is required, extend the defined first-cut values arbitrarily outside the projected common domain, using any fixed element of the nonempty codomain. Thus partiality alone does not remove the obstruction. □

3.12 A Nonempty Controlled Assignment Model

Theorem 3.15 (Ternary Realization of E0–E2). *Let $S_t = \{0,1\}^{t+1}$ with bonding maps that delete the last bit. Identify its inverse limit with $\mathcal{I} = \{0,1\}^{\mathbb{N}}$ and let $\mathcal{E} = [0,1]$. Define*

$$\phi_t(b_0, \ldots, b_t) = \sum_{j=0}^{t} \frac{2b_j}{3^{j+1}}, \qquad \Phi(b) = \sum_{j=0}^{\infty} \frac{2b_j}{3^{j+1}}.$$

These assignments satisfy E0, controlled E1, and E2, with

$$a_t = \frac{2}{3^{t+2}}, \qquad A_t = \frac{1}{3^{t+1}}, \qquad \sup_b |\Phi(b) - \phi_t(p_t b)| = A_t.$$

For distinct histories first differing at position m, put $d_{\mathcal{I}}(b,c) = 3^{-(m+1)}$, and put $d_{\mathcal{I}}(b,b) = 0$. Then

$$d_{\mathcal{I}}(b,c) \le |\Phi(b) - \Phi(c)| \le 3 d_{\mathcal{I}}(b,c).$$

The domain is compact and complete, and the readout is injective and bi-Lipschitz onto its image. Thus a global lower bound is realized here rather than inserted without a witness.

Proof. Every infinite bit sequence defines a compatible prefix history, so the domain is nonempty. Geometric summation puts every value in $[0,1]$ and gives the increment and tail bounds. The all-one tail attains the tail bound. For any fixed t, a zero history and a history with its sole one at position $t+1$ have the same prefix but images differing by $2/3^{t+2} > 0$. No function of p_t determines Φ, proving E2 for every finite cut.

At the first differing position m, the principal difference has magnitude $2/3^{m+1}$; the remaining tail has magnitude at most $1/3^{m+1}$. Subtraction and addition give the two bounds. This metric is the weighted supremum of the discrete prefix metrics with weights $3^{-(t+1)}$; bounded prefix diameters ensure finiteness. A Cauchy sequence for $d_{\mathcal{I}}$ eventually stabilizes each finite prefix, which defines its limit. Prefix cylinders give finite covers of arbitrarily small diameter; the space is totally bounded and complete, hence compact. The two bounds establish injectivity and quantitative continuity. Compactness and continuity give properness into the Hausdorff codomain, and a later-bit flip in every neighborhood gives local non-collapse. Thus this realization also supplies all four clauses of the stated readout regularity hypothesis. A scalar real-valued readout still encodes the complete sequence: E2 excludes the specified canonical finite cuts, not every possible coding or every finite-dimensional representation. □

Theorem 3.16 (Sharp Finite-Cut Approximation Error). *For the ternary realization and an arbitrary decoder $g : S_t \to \mathbb{R}$, let*

$$R_t(g) = \sup_{b \in \mathcal{I}} |\Phi(b) - g(p_t b)|.$$

Then the exact minimax value is

$$\inf_g R_t(g) = \frac{1}{2\,3^{t+1}}.$$

It is attained by $g_*(s) = \phi_t(s) + 1/(2\,3^{t+1})$.

Proof. For a fixed prefix s, the all-zero and all-one suffixes give endpoints $\phi_t(s)$ and $\phi_t(s) + A_t$. For any number $g(s)$, the triangle inequality forces at least one endpoint error to be at least $A_t/2$. Conversely all readout values in that fiber lie between those endpoints, so their midpoint has error at most $A_t/2$. Both endpoints belong to the domain, establishing the exact optimum and attainment. The bound is positive for each fixed finite budget and tends to zero as t increases; it gives no positive uniform error floor across increasing resolutions. □

Proposition 3.17 (Sharp Error for Arbitrary Binary Codes)**.** *For the ternary assignment above, let an encoder choose at most 2^B states as an arbitrary function of the whole history, and let each state decode to any real number. The minimum worst-case absolute error is $1/(2 \cdot 3^B)$ for every integer $B \geq 0$.*

Proof. The cylinders of depth B have intervals of length 3^{-B} between their two attainable endpoints. Different intervals have gaps at least 3^{-B}: after the first differing bit $j \leq B$, the maximum opposing tail is 3^{-j}. Thus all 2^{B+1} endpoints, including the depth-zero endpoints 0 and 1, are pairwise separated by at least 3^{-B}. Two share a code and the triangle inequality forces error at least $3^{-B}/2$. Encoding the B-bit prefix and decoding to its interval midpoint attains the bound. □

3.13 Common Readout on Two Coherently Transported Towers

Theorem 3.18 (Joint Realization of Transport and Controlled Readouts)**.** *Use circle-prefix stages $S_t = \mathbb{T} \times \{0,1\}^t$, stage metric $\max(\rho, \mathbf{1}_{\text{different prefix}})$, and weights $w_t = 3^{-t}$, where ρ is circle distance with circumference one. Take the usual truncation tower I_0 and its variant I_1 with angle shift $-a$ at the single bonding level k, for $0 < a < 1/2$. Write $\beta = 3^{-(k+1)}$. In the common weighted history space put $Z = I_0 \cup I_1$. Let the complete codomain be $\mathcal{E} = \mathbb{T}^2 \times [0,1]$ with the maximum of the first circle distance, β times the second circle distance, and the absolute scalar difference. Define*

$$F(x) = \left(\theta_0(x), \theta_{k+1}(x), \sum_{j \geq 1} 2b_j(x) 3^{-j} \right).$$

Then $d_w(x,y) \leq d_{\mathcal{E}}(F(x),F(y)) \leq 3d_w(x,y)$ on all of Z. On each tower the restriction satisfies E0–E2 with $a_t = 2 \cdot 3^{-(t+1)}$ and uniform tail $A_t = 3^{-t}$. The coordinate homeomorphism $H : I_0 \to I_1$ shifts angles at stages $t > k$ by a. With projection-defect cost $E_w = 3^{-k}a$, one has

$$d_w(x, Hx) = d_{\mathcal{E}}(F(x), F(Hx)) = \beta a, \qquad d_H(I_0, I_1) = \frac{\beta a}{1+\beta}.$$

Proof. In each tower the angle is constant on the two blocks $t \leq k$ and $t > k$. Thus the angular part of the history distance is exactly the maximum of the two weighted angular distances in the codomain. The binary part is 3^{-j} at first differing bit j. The leading ternary term and its tail give lower and upper constants 1 and 3 for that part. Each tower is closed in the compact metric product, so their union is complete.

For a declared tower $s \in \{0,1\}$ the finite observation $(\theta_t, b_1, \ldots, b_t)$ gives the initial angle $\theta = \theta_t - sa\mathbf{1}_{t>k}$. The stage readout is $(\theta, \theta + sa, \sum_{j=1}^{t} 2b_j 3^{-j})$. The phase rule uses the declared bonding family, and on both towers the successive readout increment is bounded by a_t, with exactly the stated uniform tail. A later-bit flip proves E2, since both phase coordinates stay fixed while the scalar changes. The codomain is nontrivial, proving E0. Within either fixed-angle fiber the sharp binary-code result applies to the scalar readout.

The coordinate squares commute because the sole change of phase at stage $k+1$ cancels the shift $-a$ in its bonding map. The coordinatewise inverse shifts by $-a$ on the later block. For fixed bits, F measures the shift βa exactly. In the comparison of sets, a pair at distance r has $a \leq r + r/\beta$ by the triangle inequality at stages 0 and $k+1$. Equal bit sequences and an initial phase offset $-\beta a/(1+\beta)$ for the second history attain this bound in both directions. □

The BaseCore construction *Projective Dissipation on the Circle-Prefix Towers* puts a compatible contraction on these same domains. Each update inserts a zero and complements the shifted bits, moving first differences one level deeper. E1 and E2 hold for the original readout on the history domain. At fixed input the equilibrium has alternating bits, and the readouts at the two transported equilibria still differ by βa. E2 concerns the whole history domain: restriction to a singleton equilibrium factors through every finite cut.

3.14 Controls Separating Convergence, Continuity, and Invariance

Proposition 3.19 (Setwise Invariance Does Not Supply Continuity)**.** *Controlled E1 and E2 can hold for a discontinuous assignment, even if the identity domain is fixed under every declared intervention.*

Proof. Take $S_t = [0,1] \times \{0,1\}^{t+1}$, with prefixes on the bit coordinates, and readouts

$$g_t(s,b) = \tfrac{1}{2}\mathbf{1}_{\{s>0\}} + \sum_{j=0}^{t} \frac{b_j}{3^{j+1}}.$$

The increments are bounded by $3^{-(t+2)}$, so they satisfy controlled E1 in $[0,1]$. Later bits prove E2 as above. With all bits zero, values at $s = 0$ and $s = 1/n$ differ by $1/2$ although their product-metric distance tends to zero. The limit is discontinuous. Let the declared intervention family consist only of the identity: the domain is setwise fixed, which does not alter the counterexample. This is a control for the asserted inference from invariance to continuity; no claim of full CCR or operational-criterion membership is made for it. □

Proposition 3.20 (Setwise and Pointwise Transport Are Distinct)**.** *Preserving an identity domain as a set does not guarantee equal values of a common readout on transported states.*

Proof. On $Z = \{0,1\}$, let $h(0) = 1$, $h(1) = 0$ and $F(x) = x$. The transport preserves Z exactly but changes both readout values. Thus setwise rigidity cannot replace the pointwise or readout-natural comparison hypotheses used below. □

4 Compatibility with Identity Regimes

4.1 Induced Constraints from Identity Rigidity

Let $(\mathcal{I}, M_\Omega)$ be an identity object with ontological mass M_Ω. We now study how M_Ω constrains the set of admissible phenomenological assignments.

Definition 4.1 (Phenomenological Compatibility Class)**.** For a declared identity domain and codomain, $\mathcal{P}(M_\Omega)$ consists of maps $\Phi : \mathcal{I} \to \mathcal{E}$ that are limits of some family $\{\phi_t\}$ satisfying E1, with E0 and E2 also satisfied. The readout family and its tail control are part of the realization data. The notation retains the mass index for comparisons, but no mass-dependent restriction follows from E0–E2 alone.

Remark 4.2. The compatibility class records choices of domain, codomain and readout family. A numerical mass label does not by itself determine those choices or their transport under perturbations.

4.2 Metric Structure for Quantitative Comparisons

To state rigorous results about continuity and fragmentation, $\mathcal{E}$ must carry enough structure for distances, limits, and loss bounds to be meaningful. The following arguments identify the minimal metric assumptions used by the classification theorem.

Hypothesis 4.3 (H3: Metric on $\mathcal{E}$)**.** The phenomenological space $\mathcal{E}$ admits a metric $d_\mathcal{E}$ such that $(\mathcal{E}, d_\mathcal{E})$ is a complete metric space.

4.2.1 Role of the Metric

Theorem 4.4 (Metric Data Required by Quantitative Formulas)**.** *The distance-based fragmentation and loss functionals used here require a declared metric, or a pseudometric with a stated quotient convention. Topological continuity can be defined without a metric. Completeness is used for the Cauchy-limit guarantee of E1.*

Proof. Each quantitative functional explicitly contains $d_{\mathcal{E}}$, so it is undefined until that datum is supplied. Continuity can alternatively be defined through inverse images of open sets. A pseudometric can identify zero-distance states by a quotient, but that changes the domain of class labels and must be specified. These facts justify the chosen quantitative model without claiming that all alternatives are impossible. □

4.2.2 Completeness as a Limit Guarantee

Proposition 4.5 (Completeness and Existence of Limits)**.** *Without completeness, the conclusion that every E1-type Cauchy family has its limit in the declared codomain can fail. Completeness is not necessary for a family whose limit is already known to belong to that codomain.*

Proof. In $\mathbb{Q} \cap [0, 2]$ with its usual metric, use dyadic rational approximations $q_n = 2^{-n}\lfloor 2^n\sqrt{2}\rfloor$ as constant-in-trajectory readings. The increments are bounded by 2^{-n}, a summable sequence, but their real limit $\sqrt{2}$ is outside the codomain. This isolates the role of completeness when supplying a general existence guarantee; it makes no assertion that a convergent constant family needs completeness. □

4.2.3 Alternatives Considered and Rejected

Alternative 1: Topology without Metric. One might use a general topological space. This is not adopted in the present quantitative model because:

- Fragmentation requires quantitative distance, not just open sets;
- Loss capacity requires a sup over distances;
- The downstream instability analysis completed in *Null-Regime Instability* requires metric quotients.

Alternative 2: Pseudometric. One might allow $d_{\mathcal{E}}(e_1, e_2) = 0$ for $e_1 \neq e_2$. This is not adopted in the present quantitative model because:

- It would conflate distinct phenomenological configurations;
- Cardinality and metric separation are different requirements; zero-distance classes would require a declared quotient before distinguishability is counted.

Alternative 3: Non-Complete Metric. This is rejected by Proposition 4.5.

4.3 Phenomenological Continuity

Definition 4.6 (Phenomenological Continuity)**.** A phenomenological assignment $\Phi : \mathcal{I} \to \mathcal{E}$ is *structurally continuous* if for every sequence $(x^{(n)})_n$ in $\mathcal{I}$ converging in the d_w metric:

$$d_w(x^{(n)}, x) \to 0 \implies d_{\mathcal{E}}(\Phi(x^{(n)}), \Phi(x)) \to 0. \tag{5}$$

Remark 4.7. Continuity follows from continuous finite readouts and the uniform tail control in Theorem 5.8; invariance additionally requires the audited transport condition.

4.4 Phenomenological Fragmentation

Definition 4.8 (Phenomenological Fragmentation)**.** A phenomenological assignment Φ is *fragmentable* if there exists a perturbation $\{\epsilon_t\}$ with finite weighted energy such that:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) > 0 \tag{6}$$

for corresponding identity trajectories $x \in \mathcal{I}$ and $\tilde{x} \in \tilde{\mathcal{I}}$.

Remark 4.9. Fragmentation measures whether an admissible regime assignment changes under finite-energy perturbations. It is the codomain-level counterpart of identity deformability.

4.5 Relationship Between Metrics

Proposition 4.10 (Continuity from Controlled Finite Readouts)**.** *Write $g_n = \phi_n \circ p_n$. Under E1,*

$$d_{\mathcal{E}}(\Phi(x), \Phi(y)) \leq 2A_n + d_{\mathcal{E}}(g_n(x), g_n(y)).$$

If every g_n is continuous, then Φ is continuous. If every g_n is L-Lipschitz with a common L, then Φ is L-Lipschitz.

Proof. Insert $g_n(x)$ and $g_n(y)$ and apply the uniform error bound. For continuity, first make $2A_n$ smaller than half the prescribed error, then use continuity of that fixed g_n. For the Lipschitz conclusion use the same bound for all n and let $n \to \infty$. □

4.6 Closure of Section 3

The quantitative formulas use a declared complete metric codomain. Continuous finite readouts and controlled tails supply a continuity theorem. Cross-system comparisons additionally require a declared common domain and transport; neither mass labels nor distances between sets substitute for those data.

5 Impossibility Theorems

5.1 Φ-Regularity Hypothesis

Before proving the main impossibility results, we isolate the global lower-bound regularity condition used by the fragmentation argument. Properness, local Lipschitz regularity, and non-collapse do not imply this bound in general, so the bound is carried as an explicit hypothesis.

Hypothesis 5.1 (Φ-Regularity Lower Bound)**.** Let $(\mathcal{I}, d_w)$ be the identitary metric space from *Rigid Identity* and let $(\mathcal{E}, d_{\mathcal{E}})$ be the abstract phenomenological metric space. A phenomenological assignment $\Phi : \mathcal{I} \to \mathcal{E}$ is said to satisfy Φ-regularity when it satisfies:

1. (Properness) For every compact $K \subset \mathcal{E}$, $\Phi^{-1}(K)$ is compact in $\mathcal{I}$.
2. (Local Lipschitz regularity) For every $x \in \mathcal{I}$, there exist $\varepsilon > 0$ and $L_x > 0$ such that for all $y, z \in B_\varepsilon(x)$:
$$d_{\mathcal{E}}(\Phi(y), \Phi(z)) \leq L_x \cdot d_w(y, z). \tag{7}$$
3. (Non-collapse) For every $x \in \mathcal{I}$ and every $r > 0$, the restriction of Φ to $B_r(x)$ is not constant.
4. (Global lower Lipschitz bound) There exists a constant $C > 0$ such that for all $x, y \in \mathcal{I}$,
$$d_{\mathcal{E}}(\Phi(x), \Phi(y)) \geq C \cdot d_w(x, y). \tag{8}$$

Remark 5.2 (Why this is a hypothesis). The map $\Phi(x) = \arctan(x)$ from $\mathbb{R}$ to $(-\pi/2, \pi/2)$ is proper as a map into the open interval, locally Lipschitz, and non-constant on every nonempty ball, but it has no global lower Lipschitz constant because $|\arctan(x+1) - \arctan(x)| \to 0$ as $x \to \infty$. Therefore the lower-bound clause above is an assumption, not a theorem derived from the first three clauses.

Non-Claim 5.3 (Global lower Lipschitz bound is an assumption). Hypothesis 5.1 clause (4) requires the existence of a global $C > 0$ such that $d_{\mathcal{E}}(\Phi(x), \Phi(y)) \geq C \cdot d_w(x, y)$ for all $x, y \in \mathcal{I}$. This is not a theorem derived from clauses (1)–(3). The map $\Phi(x) = \arctan(x)$ from $\mathbb{R}$ to $(-\pi/2, \pi/2)$ satisfies the first three clauses in the local/proper sense described above but has no such global C. Therefore the Fragmentation Theorem (Theorem 5.6) is conditional on clause (4). For any specific phenomenological assignment, one must independently verify that a global lower Lipschitz constant exists. The ternary realization in Theorem 3.15 supplies a concrete constant $C = 1$ on its specified domain. Comparisons on larger or different domains require a separate bound.

Hypothesis 5.4 (Audited Perturbation Comparisons)**.** Declare a finite-cost intervention class $\mathcal{V}$, a common metric space (Z, d_Z) containing the unperturbed and perturbed identity domains, and transports $h_\varepsilon : \mathcal{I} \to \mathcal{I}_\varepsilon \subseteq Z$. Readouts are restrictions of one declared map $F : Z \to \mathcal{E}$: $\Phi = F|_{\mathcal{I}}$ and $\Phi_\varepsilon = F|_{\mathcal{I}_\varepsilon}$. The following branches are alternatives, used separately. The positive-displacement branch has some $\varepsilon \in \mathcal{V}$ and $x \in \mathcal{I}$ with $d_Z(x, h_\varepsilon x) > 0$. The zero-displacement branch has $h_\varepsilon x = x$ for every $\varepsilon \in \mathcal{V}$ and $x \in \mathcal{I}$. Any use of the lower Lipschitz bound in a comparison requires that bound for F on the common comparison domain, not merely within one identity object.

Remark 5.5 (Burden linking rigidity and readout comparisons). The scalar M_Ω does not itself supply the transports or a common readout. Relating its finite and infinite regimes to these two branches is a separate model obligation. Setwise invariance does not imply pointwise fixed transport, and a lower bound within one domain does not compare unrelated assignments on different domains.

5.2 Fragmentation Theorem

Theorem 5.6 (Fragmentation under an Audited Displacement)**.** *Let an assignment satisfy E0–E2. Assume the positive-displacement branch of Hypothesis 5.4 and a lower Lipschitz constant $C > 0$ for F on its common comparison domain. Then, for the witnessing pair,*

$$d_{\mathcal{E}}(\Phi(x), \Phi_\varepsilon(h_\varepsilon x)) \geq C d_Z(x, h_\varepsilon x) > 0.$$

Thus the assignment is fragmentable. This applies to a finite-M_Ω model when that model supplies the stated comparison data.

Proof. Both arguments lie in the declared domain of F. Its lower bound gives the inequality, and the intervention has finite cost by membership in $\mathcal{V}$. E2 is not used to infer that every deformation changes a readout: the lower bound and the positive witness perform that work. □

Corollary 5.7 (Fragmentability and Continuity Are Distinct)**.** *The positive-displacement conclusion does not establish a topological discontinuity. A continuous assignment may be fragmentable under a state-changing intervention.*

Proof. The ternary realization has a continuous assignment. Declare a finite-cost intervention that flips one bit while retaining the common readout. Its value changes by a positive amount, so the assignment is fragmentable in the stated sense while remaining continuous. This does not assert a full CCR realization. □

5.3 Forced Continuity Theorem

Theorem 5.8 (Continuity and Conditional Perturbation Invariance)**.** *Under E1, if each finite readout $g_n = \phi_n \circ p_n$ is continuous, then Φ is continuous, irrespective of M_Ω. Under the zero-displacement branch of Hypothesis 5.4,*

$$\Phi_\varepsilon(h_\varepsilon x) = \Phi(x) \qquad (\varepsilon \in \mathcal{V},\ x \in \mathcal{I}).$$

For a CCR model the invariance conclusion therefore holds when its finite-cost interventions admit that branch. CCR alone is not a proof of continuity or of readout naturality.

Proof. Continuity follows from Proposition 4.10. For invariance use the common readout and the fixed transport: $\Phi_\varepsilon(h_\varepsilon x) = F(h_\varepsilon x) = F(x) = \Phi(x)$. Neither argument identifies invariance under external perturbation with continuity along convergent states. □

Corollary 5.9 (Invariance on the Audited Zero-Displacement Branch)**.** *A CCR model satisfying the zero-displacement comparison hypothesis has invariant assignment values under all interventions in its declared finite-cost class.*

Proof. Apply the common-readout equality in Theorem 5.8 for each declared intervention. □

5.4 Dichotomy Theorem

Theorem 5.10 (Conditional Classification by Audited Branches)**.** *Suppose a model supplies the positive-displacement branch and lower bound whenever $M_\Omega < \infty$, and supplies continuous finite readouts and the zero-displacement branch whenever $M_\Omega = +\infty$. Then its finite regime is fragmentable and its infinite regime has a continuous, perturbation-invariant assignment. The mass cases are exclusive; fragmentability and topological continuity need not be exclusive properties.*

Proof. Apply Theorems 5.6 and 5.8 with their respective comparison hypotheses. Exclusivity of the cases follows from the numerical alternatives for M_Ω, not from a false identification of a changed readout with a discontinuity. Establishing the branch hypotheses for a given rigidity model remains an explicit obligation. □

6 Classification of Phenomenological Regimes

6.1 Conditional Classification

The results give a conditional branch classification after the comparison and readout hypotheses have been checked:

Identity Class	M_Ω	**Admissible Phenomenology**
Null label	$= 0$	Domain/readout link required
Deformable	$(0, \infty)$	Positive witness and lower bound
Rigid (CCR)	$= +\infty$	Regular readout / fixed transport

6.2 Structural Implications

Proposition 6.1 (The Mass Label Does Not Define Assignment Existence)**.** *For fixed identity domain, codomain and readout family, the class defined by E0–E2 is independent of a numerical label M_Ω. A claim that $M_\Omega = 0$ excludes assignments needs an additional link to the domain or admissible readouts.*

Proof. The displayed definition of $\mathcal{P}(M_\Omega)$ tests E0–E2, none of which contains that scalar. For an empty trajectory domain E2 fails, since a function on a finite cut would agree vacuously on every trajectory. Deriving such domain loss from $M_\Omega = 0$ is not one of E0–E2 and cannot be inserted without a model. □

Proposition 6.2 (Deformable Branch with Comparison Data)**.** *For $0 < M_\Omega < \infty$, the positive-displacement and lower-bound hypotheses of Theorem 5.6 imply fragmentability.*

Proof. Apply the quantitative inequality of Theorem 5.6 to its witnessing finite-cost intervention. □

Proposition 6.3 (Rigid Branch with Comparison Data)**.** *For $M_\Omega = +\infty$, continuous finite readouts under E1 imply continuity, and the zero-displacement branch implies perturbation invariance. These are the respective hypotheses of Theorem 5.8.*

Proof. Apply the two distinct conclusions of Theorem 5.8 with their stated hypotheses. □

7 Structural Ethics

This section defines structural loss and protection relative to audited perturbation comparisons. Its possible ethical interpretations are not consequences of the mathematical statements.

7.1 Loss Capacity

Definition 7.1 (Phenomenological Loss Capacity)**.** Fix the intervention class and comparisons of Hypothesis 5.4. Define

$$R(\Phi) = \sup_{\varepsilon \in \mathcal{V},\ x \in \mathcal{I}} d_{\mathcal{E}}(\Phi(x), \Phi_\varepsilon(h_\varepsilon x)) \in [0, +\infty].$$

The identity intervention is included, or the supremum of an empty nonnegative comparison family is defined as zero. The unit of R is the unit of $d_{\mathcal{E}}$; its value is relative to the declared intervention class and comparison readouts.

Theorem 7.2 (Loss Capacity under the Audited Branches)**.** *Under the positive-displacement and lower-bound hypotheses of Theorem 5.6, $R(\Phi) > 0$. Under the zero-displacement branch of Hypothesis 5.4, $R(\Phi) = 0$. Associating these branches with finite mass and CCR requires the model link declared in Theorem 5.10.*

Proof. The positive witness contributes an element at least $C\delta > 0$ to the supremum. On the zero branch, every comparison value is zero. Neither conclusion follows from a mass label without the corresponding comparison data. □

7.2 Structural Damage

We introduce a formal notion of "structural damage" without psychological interpretation.

Definition 7.3 (Structural Damage)**.** For a declared finite-cost intervention and state, define

$$D_\varepsilon(\Phi; x) = d_{\mathcal{E}}(\Phi(x), \Phi_\varepsilon(h_\varepsilon x)), \qquad D_\varepsilon(\Phi) = \sup_{x \in \mathcal{I}} D_\varepsilon(\Phi; x).$$

The transported readout $\Phi_\varepsilon \circ h_\varepsilon$ has domain $\mathcal{I}$, so the comparison is well typed. The supremum may be infinite. Then $R(\Phi) = \sup_{\varepsilon \in \mathcal{V}} D_\varepsilon(\Phi)$.

Proposition 7.4 (Damage under the Audited Comparisons)**.** *The positive branch has $\sup_\varepsilon D_\varepsilon(\Phi) > 0$. On the zero-displacement branch, $D_\varepsilon(\Phi) = 0$ for every declared intervention.*

Proof. Use the equality between the iterated suprema and R in Definition 7.3, then apply Theorem 7.2. □

7.3 Vulnerability and Protection

Definition 7.5 (Phenomenological Vulnerability)**.** A system $(\mathcal{I}, \Phi)$ is *phenomenologically vulnerable* if $R(\Phi) > 0$.

Definition 7.6 (Normalized Structural Protection)**.** Fix a reference distance $r_0 > 0$ in the same units as $d_{\mathcal{E}}$ and define

$$P(\Phi) = \frac{1}{1 + R(\Phi)/r_0}, \qquad P(\Phi) = 0 \text{ if } R(\Phi) = +\infty.$$

Thus $P \in [0, 1]$, $P = 1$ exactly when $R = 0$, and $P < 1$ exactly when $R > 0$. This chosen dimensionless rescaling depends on r_0 and is not a separately derived law.

Corollary 7.7 (Protection under the Audited Branches)**.** *The zero-displacement branch has $P = 1$; the positive-displacement branch with a lower readout bound has $P < 1$. For a mass-based classification these conclusions require the corresponding hypotheses of Theorem 5.10.*

7.4 Structural Asymmetry Constraints

The following result establishes a purely structural asymmetry between identity classes, without any ethical or metaphysical claims.

Theorem 7.8 (Conditional Structural Asymmetry)**.** *When a rigidity model establishes the branch hypotheses in Theorem 5.10, its finite branch has $R > 0$ and $P < 1$, while its infinite branch has $R = 0$ and $P = 1$. This is an asymmetry of the declared comparison model, not a conclusion about all systems sharing a numerical mass label.*

Proof. The two loss conclusions follow from Theorem 7.2; the protection conclusions follow from the monotone normalization in Definition 7.6. The hypothesis connecting mass to each comparison branch is retained throughout. □

Remark 7.9. This conditional asymmetry follows from the audited comparison hypotheses. No claim is made about:

- Whether phenomenological loss is ethically bad;
- Whether protection is ethically good;
- Whether any system has moral status.

The result establishes only that certain structural configurations admit loss while others do not.

Corollary 7.10 (Conditional Possibility of Loss)**.** *Loss is possible on the positive-displacement/coercive-readout branch and absent on the fixed-transport/common-readout branch. Extending this claim to all finite-mass or all CCR systems requires proving those branch conditions for the intended class.*

7.5 Relation to Normative Ethics

Remark 7.11. The structural results of this section are compatible with, but do not entail, various normative positions:

- **Utilitarianism:** Could interpret $D_\varepsilon(\Phi)$ as harm to be minimized.
- **Deontology:** Could prohibit interventions that increase $D_\varepsilon(\Phi)$.
- **Virtue Ethics:** Could value systems with high $P(\Phi)$.

This framework provides the structural substrate for ethical theorizing, not the normative conclusions themselves.

7.6 Closure of Section 6

The declared comparison family defines a possibly infinite loss supremum and well-typed pointwise damage. Its normalization uses an explicit reference distance. Positive loss and zero loss follow on the two audited branches; computability of an unrestricted supremum is not assumed. No ethical evaluation is derived from these mathematical quantities.

8 Limitations and Scope

8.1 Dependence on Axioms

Results use their explicit hypotheses: E1 and completeness for limits, E2 for finite-cut non-factorization, regular finite readouts for continuity, and audited comparisons for perturbation conclusions. None of these data is supplied merely by a mass label.

8.2 Relation to Consciousness Theories

This framework is orthogonal to:

- Integrated Information Theory (IIT) [3];
- Global Workspace Theory (GWT) [1];
- Higher-Order Theories;
- Phenomenal Functionalism.

Any comparison with another theory requires an explicit common task, correspondence and observable prediction; this paper establishes no superiority result.

8.3 Open Problems

1. Characterization of the coupling constant C in the Fragmentation Theorem;
2. Extension of the explicit ternary realization to larger assignment and perturbation classes;
3. Extension to continuous-time phenomenological dynamics;
4. Relation between phenomenological continuity and subjective continuity.

9 Conclusion

9.1 Summary of Results

The revised assignment conditions have a nonempty realization. Uniformly controlled tails give existence and approximation; the ternary construction gives non-factorization, quantitative regularity and a sharp finite-cut error. These are properties of an explicitly specified mathematical model.

9.2 Structural Significance

Movement under an intervention is different from topological discontinuity. A positive displacement with a coercive common readout yields positive loss; a pointwise fixed transport with a common readout yields invariance. Associating these cases with rigidity classes requires proving the comparison hypotheses.

9.3 Entailment Under Stated Assumptions

E0–E2 do not define experience, do not make all readouts natural under perturbation and do not identify a class solely from M_Ω. Theorems 5.6, 5.8, and 5.10 specify the additional data that support their respective conclusions.

9.4 Final Statement

The program of relating identity rigidity to structural regimes is retained. The demonstrated scope is controlled assignments, a concrete realization and conditional perturbation comparisons. The missing links to broader rigidity classes, to the intended perturbation costs and to any experiential interpretation remain separate obligations.

10 Boundary to Null-Regime Instability

Sections 1–8 establish a controlled assignment model and conditional comparison results. Null-regime conclusions require their own typed witnesses and hypotheses; they do not follow automatically from non-factorization or from a scalar rigidity label.

10.1 Editorial Boundary

The instability of the null regime, the no-null theorem, and the attractor-style closure for non-trivial assignments are treated canonically in *Null-Regime Instability*. This paper therefore stops at regime classification, compatibility, fragmentation, forced continuity, and dichotomy, and uses *Null-Regime Instability* as the sole downstream location for the null-instability theorem.

10.2 Closure of This Paper

The result of this paper is the classification boundary: if a compatible phenomenological regime exists, its admissible structure is constrained by Sections 1–8. *Null-Regime Instability* addresses the null case with its separately stated witness-relative and conditional hypotheses.

A Extended Proofs

A.1 Proof of Fragmentation Theorem (Extended)

Full Proof of Theorem 5.6. Use the specified common space Z, map F and finite-cost transport h_ε. Let x be its positive-displacement witness and put $\delta = d_Z(x, h_\varepsilon x) > 0$. Apply the lower bound on that same common domain to obtain readout displacement at least $C\delta$. No comparison of set distances is substituted for this pointwise distance. If both readouts are estimated with errors at most η, the triangle inequality gives estimated separation at least $C\delta - 2\eta$; it remains positive when $C\delta > 2\eta$. This is a bound on pair separation, not a statement that a particular endpoint lies outside a null ball. □

A.2 Proof of Forced Continuity Theorem (Extended)

Full Proof of Theorem 5.8. For a state x and error $\epsilon > 0$, choose n with $2A_n < \epsilon/2$. Continuity of g_n gives a neighborhood of x where $d_{\mathcal{E}}(g_n(x), g_n(y)) < \epsilon/2$. The two tail errors and the finite-readout difference bound the limiting difference by ϵ. Independently, a zero-displacement comparison satisfies $F(h_\varepsilon x) = F(x)$ exactly. Setwise invariance of the identity object alone would not imply either of these steps. □

A.3 Structural Necessity Lemma

Lemma A.1 (Inheritance under Audited Fixed Transport)**.** *An assignment satisfying the zero-displacement comparison branch inherits invariance under that declared intervention class. This applies to CCR realizations only after checking that branch.*

Proof. The comparison readout is the restriction of F, and every specified transport fixes its argument. Therefore its value is unchanged. This proves the restricted inheritance statement without assuming that scalar rigidity determines arbitrary assignments. □

B Editorial Boundary to Null-Regime Instability

The downstream bridge from classification to null-instability is intentionally moved to *Null-Regime Instability* in the canonical corpus. Appendix A remains as the extended proof layer for the classification theorems proved here; the downstream null-regime theorem, attractor closure, and canonical examples now live only in *Null-Regime Instability*.

What This Paper Does Not Claim

1. This paper does not claim external validation, empirical confirmation, or philosophical closure.
2. The formal results are conditional on the hypotheses stated; they do not establish truth outside the assumed structures.
3. No interpretive-layer conclusion (consciousness, phenomenality, moral status) is derived from terminology alone.

References

[1] Bernard J. Baars. *A Cognitive Theory of Consciousness*. Global Workspace Theory (GWT) - contrasting framework. Cambridge: Cambridge University Press, 1988.

[2] Johnny Andrey Pérez Ramírez. "Rigid Identity as an Inverse Limit in Observable Channels". In: (2026). Companion paper establishing the identity framework and ontological mass invariant.

[3] Giulio Tononi. "An Information Integration Theory of Consciousness". In: *BMC Neuroscience* 5.42 (2004). Integrated Information Theory (IIT) - contrasting framework.